1. LaTeX

a. Basic use

Accessing a single palette color is straightforward: use \palUse{<name>}{<index>} where <name> is the standard palette name (without prefix), and <index> is the color number. For example, \palUse{Accent}{8} is the eighth color of the Accent palette, an xcolor format color that can be easily used as shown in the following compilable example.

```
\documentclass{article}

% Load the wanted palette.
\usepackage{palettes-hf/Accent}

% Load the palette API.
\usepackage{palapi}

\usepackage{tikz}

\begin{document}

\textcolor{\palUse{Accent}{8}}{\bfseries Colored text.}

Representation of the first ten palette colors.

\begin{tikzpicture}
  \foreach \i in {1,...,10} {
    \fill[\palUse{Accent}{\i}]
      (1.25*\i - 1, 0) rectangle (1.25*\i, 1);
  }
\end{tikzpicture}

\end{document}
```

b. Creating palettes from scratch

Internally, palettes are defined using an array-of-three-floats syntax. For instance, the following snippet is an excerpt from palettes-hf/Accent.sty.

```
\palCreateFromRGB{Accent}{
  {0.4980392157, 0.7882352941, 0.4980392157},
  {0.7450980392, 0.6823529412, 0.831372549},
  ...
}
```

For creating new palettes manually, the following high-level commands are available.

1. \palCreateFromNames works with a comma separated list of named colors, while \palCreateFromRGB creates a palette by entering it as an array-like variable of arrays of three floats.

2. \palSize{<name>} returns the palette size (useful for loops, for example).

The following example demonstrates the \palCreateFromRGB and \palCreateFromNames commands.

```
\usepackage{palapi}
\usepackage[svgnames]{xcolor}

\palCreateFromRGB{MyRGBPal}{
  {0.0, 0.0, 0.0},
  {0.8, 0.2, 0.0},
}

\palCreateFromNames{MyNameUsePal}{
  YellowGreen,
  green!60!black,
}
```

Note.

All built-in palettes are created using the `\palCreateFromRGB` *macro.*

A lower-level approach is also available through the following commands.

1. `\palNew{<name>}` initializes a new (empty) palette.
2. `\palAddNames{<name>}{<color-using-names>}` appends a color defined with named colors to the palette.
3. `\palAddRGB{<name>}{<r>, <g>, <b>}` appends an RGB color to the palette, where `<r>`, `<g>`, and `<b>` are decimal values ranging from 0 to 1.

The following example demonstrates the flexibility offered by these low-level commands.

```
\usepackage{palapi}
\usepackage[svgnames]{xcolor}

\palNew{LowLevelPal}
\palAddNames{LowLevelPal}{IndianRed}
\palAddRGB{LowLevelPal}{0.0, 0.0, 0.0}
\palAddNames{LowLevelPal}{green!60!black}
\palAddRGB{LowLevelPal}{0.8, 0.2, 0.0}
```

c. Creating palettes from existing ones

Building new palettes by transforming existing ones can be achieved using the `\palCreateFromPal` command, which has the signature `\palCreateFromPal{<new-name>}[<options>]{<existing-name>}`. The following example shows how to do this (all options are used).

```
\usepackage{palettes-hf/Accent}
\usepackage{palapi}

\palCreateFromPal{AccentTransformed}[
  extract = {1, 3, 6, 9},
  shift   = 1,
  reverse
]{Accent}
```

Tip.

`\palCreateFromPal{<new-name>}{<existing-name>}` *builds a copy of an existing palette.*

d. Retrieving the internal definition of a palette

The internally stored definition of a palette named `MyPal`, for example, is `\g_palette_MyPal_seq` which is a LaTeX3 variable (keep in mind the pattern `\g_palette_<palette-name>_seq`).

Note.

Variables of type `\g_palette_<palette-name>_seq` *are not used internally to retrieve the colors themselves; they are only there for technical reasons related to the development process of new palettes via LaTeX.*